

CHULMAN NERYUNGRI, Russian Federation

WMO: 303930

Lat: 56.914N

Lon: 124.912E

Elev: 858

StdP: 91.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.2	(c) -40.2	(d) -44.2	(e) 0.1	(f) -42.1	(g) -42.2	(h) 0.1	(i) -40.0	(j) 7.8	(k) -28.0	(l) 7.1	(m) -28.7	(n) 0.6	(o) 70	(p) 0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 26.9	(d) 16.3	(e) 25.0	(f) 15.9	(g) 23.1	(h) 15.0	(i) 18.0	(j) 23.4	(k) 17.1	(l) 22.3	(m) 16.2	(n) 21.1	(o) 3.4	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	12.9	19.3	15.3	12.1	18.6	14.2	11.2	18.0	54.6	23.4	51.6	22.3	48.8	21.1	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.7	(b) 6.7	(c) 6.1	(e) -44.4	(f) 29.6	(g) 2.6	(h) 2.2	(i) -46.3	(j) 31.2	(k) -47.8	(l) 32.5	(m) -49.3	(n) 33.7	(o) -51.2	(p) 35.3		
			DB													
			WB	-44.4	20.0	2.5	1.2	-46.2	20.8	-47.7	21.5	-49.1	22.1	-51.0	23.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.5	-30.3	-24.7	-15.5	-4.7	4.9	14.0	16.6	13.5	4.8	-6.9	-21.1	-30.1
	DBStd	17.63	6.23	5.78	6.20	4.91	4.52	4.32	3.54	4.10	4.57	5.80	7.86	6.39
	HDD10.0	6518	1249	972	789	442	167	15	2	16	164	524	934	1244
	HDD18.3	9110	1507	1205	1048	692	416	137	75	156	406	782	1184	1502
	CDD10.0	484	0	0	0	0	9	137	206	123	8	0	0	0
	CDD18.3	34	0	0	0	0	0	8	21	5	0	0	0	0
	CDH23.3	365	0	0	0	0	3	107	190	65	1	0	0	0
	CDH26.7	60	0	0	0	0	0	17	37	7	0	0	0	0
Wind	WSAvg	2.8	1.9	2.4	3.0	3.4	3.5	3.1	2.8	2.9	3.1	3.2	2.5	1.9
Precipitation	PrecAvg	530	15	10	14	29	47	87	104	91	62	36	22	17
	PrecMax	659	30	20	45	59	112	189	240	189	145	68	68	30
	PrecMin	364	4	5	4	10	11	13	27	24	18	14	7	8
	PrecStd	74	6	4	9	13	25	36	41	37	24	14	11	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.8	-8.0	1.6	11.2	22.3	28.8	30.1	27.5	20.9	9.0	-0.1	-11.8
		MCWB	-12.1	-9.2	-1.9	4.1	11.5	16.3	17.6	16.8	12.4	4.3	-2.3	-12.1
	2%	DB	-15.6	-10.8	-1.5	7.3	18.2	26.2	27.3	25.2	17.1	5.4	-3.3	-15.1
		MCWB	-15.8	-11.6	-4.0	2.1	9.4	15.4	16.9	16.2	10.7	1.9	-4.7	-15.3
	5%	DB	-18.9	-13.4	-3.9	4.9	15.5	24.1	25.4	23.0	14.6	2.7	-6.4	-18.0
		MCWB	-19.1	-14.0	-6.0	0.6	7.9	14.2	16.7	15.5	9.7	0.3	-7.3	-18.3
	10%	DB	-21.2	-16.1	-6.0	2.9	12.9	22.1	23.5	20.8	12.1	0.7	-9.5	-20.8
		MCWB	-21.4	-16.6	-7.5	-0.6	6.6	13.3	16.0	14.5	8.4	-1.0	-10.1	-20.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.9	-9.3	-1.5	5.0	12.2	18.0	19.6	18.7	13.7	4.9	-2.0	-12.2
		MCDB	-11.8	-8.2	1.2	9.7	20.6	23.7	24.8	23.6	18.1	8.4	-0.8	-12.1
	2%	WB	-15.8	-11.6	-4.0	2.5	10.1	16.5	18.5	17.6	12.0	2.3	-4.7	-15.4
		MCDB	-15.6	-10.8	-1.5	6.6	16.6	23.4	23.9	22.5	15.0	4.9	-3.4	-15.2
	5%	WB	-19.1	-14.1	-5.8	0.9	8.6	15.4	17.6	16.5	10.3	0.5	-7.4	-18.4
		MCDB	-18.9	-13.5	-4.1	4.2	14.1	22.1	22.9	21.1	13.6	2.2	-6.4	-18.2
	10%	WB	-21.5	-16.6	-7.6	-0.4	7.2	14.4	16.8	15.4	8.7	-0.9	-10.3	-20.9
		MCDB	-21.3	-16.1	-6.0	2.5	11.8	20.4	21.7	19.6	11.7	0.5	-9.7	-20.7
Mean Daily Temperature Range	5% DB	MDBR	6.6	8.3	9.9	9.5	9.5	11.2	10.3	10.4	8.8	7.3	6.9	6.3
		MCDBR	7.9	9.7	10.9	11.2	14.3	14.5	13.6	14.0	13.7	9.4	7.8	7.9
		MCWBR	7.7	8.9	8.8	7.2	7.6	6.6	5.9	6.6	8.2	6.4	6.8	7.8
	5% WB	MCDBR	7.8	9.6	10.6	10.2	12.7	12.2	11.0	11.5	11.6	8.2	7.8	7.8
		MCWBR	7.6	8.9	8.6	6.8	7.4	6.5	5.7	6.2	7.4	5.9	6.9	7.8
Clear-Sky Solar Irradiance	taub		0.232	0.263	0.294	0.381	0.442	0.415	0.438	0.385	0.327	0.289	0.241	0.221
	taud		2.069	2.085	2.065	1.897	1.991	2.244	2.192	2.339	2.442	2.266	2.074	1.999
	Ebn at Noon		651	785	853	816	792	826	794	819	826	750	621	540
	Edh at Noon		57	87	117	165	161	127	132	106	82	73	55	45
All-Sky Solar Radiation	RadAvg		0.84	1.91	3.72	5.50	5.30	5.38	4.85	3.93	2.69	1.64	1.03	0.57
	RadStd		0.03	0.09	0.14	0.25	0.49	0.41	0.51	0.40	0.24	0.16	0.07	0.02

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.44	N/A	+1.24	N/A	N/A	N/A	N/A	-165	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page